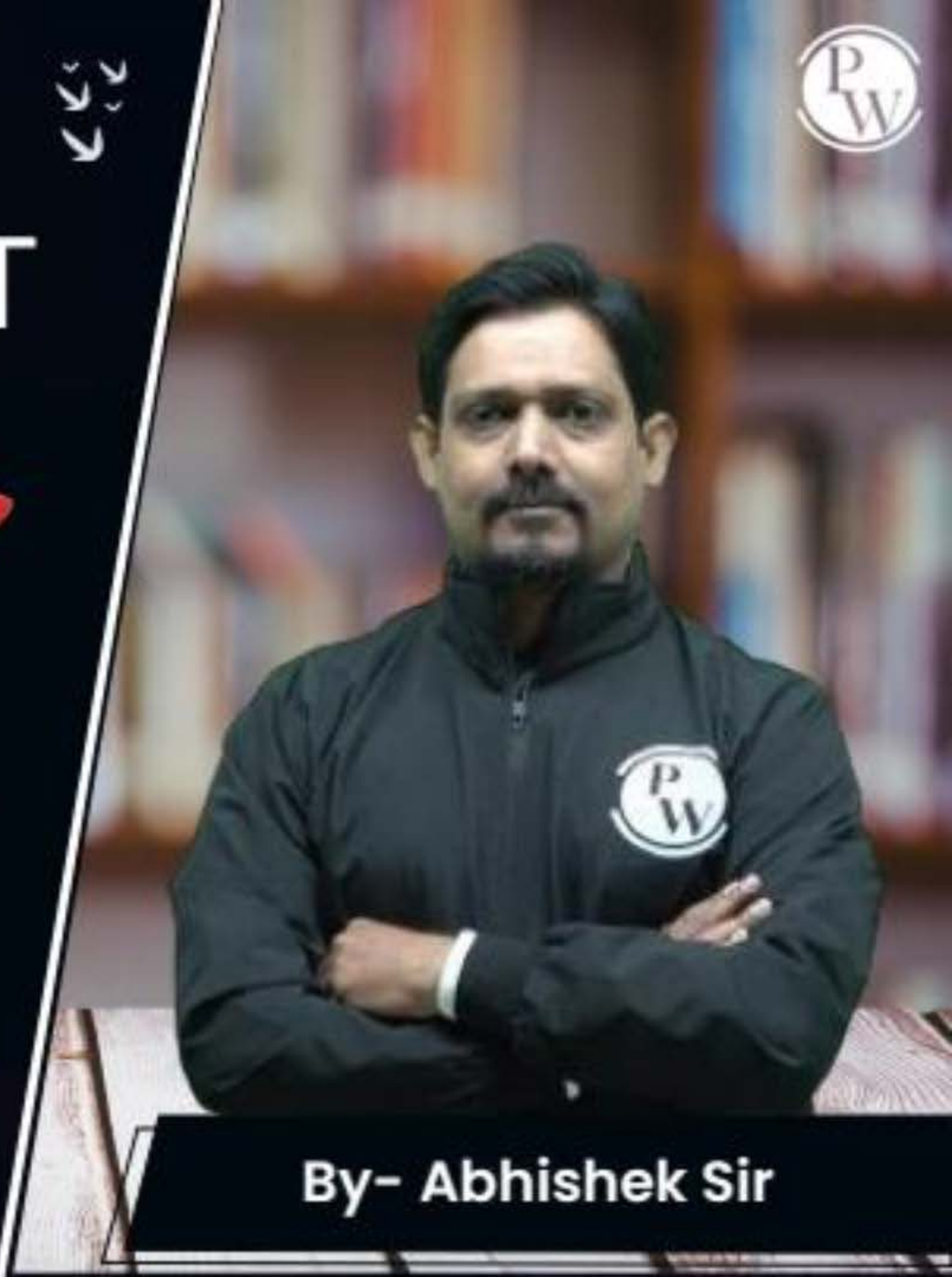
Computer Science & IT



Fundamental of C Language

Data Types & Operators

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



- topic
- topic
- topic
- topic
- topic

Data type

int unsigned. Signed

Tokens : keywords, identifier
operator. Numeric constant
special character

Signed a = 10;

Slide

topics to be Covered



- topic
- topic
- topic
- topic
- topic

short long.

char

floating point

Slide



Topic : Integer Datatype

short 2 long integer standard size : 4B

short int a = 10; a Size = 2Byte

printf("%d", a); // this statement print 10

short a = 10;

Slide



Topic : Integer Datatype

Short & long

long int a = 10;

long a = 10; ?

printf ("%ld", a);

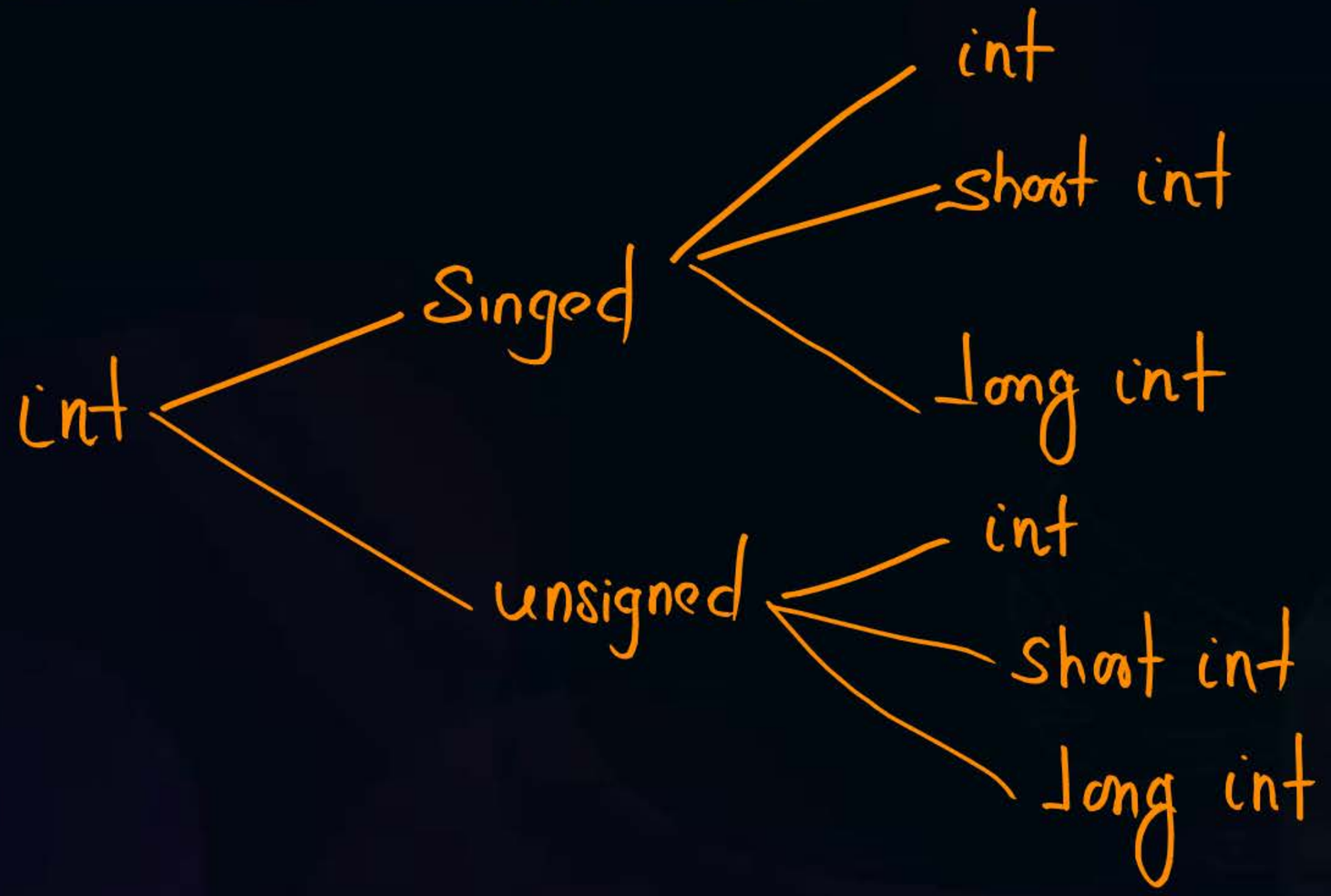
long integer Size = 8 Byte

8 x 8 = 64 bits

Slide



Topic : Integer Datatype



long long int

long : 8B

long long : 8B

Suppose

long : 4B

long long : 8B

Slide



Topic : Char Data Types

```
char ch = 'a';
```

keyword identifier operator character constant

Character constant in single quotes

Slide



Topic : Char Data Types

Size of character is 1 Byte ≡ 8 bits

```
char ch = 'a';

printf("%c", ch); // output  a
```

Slide



Topic : Char Data Types

character in c language is represented by ASCII value

American Standard code for information interchange"

ASCII table : for every character there is a Numeric value.

Slide



Topic : Char Data Types

A	65
B	66
C	67
D	68
E	69
:	
Z	90

a	97
b	98
c	99
d	100
:	
z	122

0	48
1	49
2	50
3	51
4	52
5	53
6	54
7	55
8	56
9	57

Slide



Topic : Char Data Types

8 bits

A - 65 $\xrightarrow{\text{Binary value}}$ 01000001

with every character a decimal value is associated

Slide



Topic : Char Data Types

```
char ch = 'a';
printf("%c", ch);  ⟶ a printed
printf("%d", ch);  ⟶ ASCII value // 97
ch = ch+4;  ← correct statement "No error"
printf("%c", ch); // ch new value is 97+4 = 101
                  e print
```

97 - a
98 - b
99 - c
100 - d
101 - e

Slide



Topic : Char Data Types

ASCII table



Topic : Float Data Types

Real No float data type

```
float pi = 22/7;
printf ("%f", pi);

double pi1 = 27/7;
printf ("%lf", pi1);
```

float is 4 Byte

4 × 8 = 32 bits

double is 8 Bytes

8 × 8 = 64 bits



Topic : Float Data Types

Floating point standard is called IEEE-754

← 32 bits



Slide



2 mins Summary

topic

Short , long

topic

Char data type

topic

float data type

topic

topic

Slide

THANK - YOU